

ActInf Textbook Group ~ Cohort 1 ~ Meeting 23

(Chapter 10, part 1)

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hello it's active textbook group cohort one meeting 23 on no 4 22 and we are in our first of two discussions on chapter 10. It's the final chapter of the book and so let's take a look at the table of contents and just does anybody from there reading or experience want to share an initial note on where and how we've come where do we sit in a satisfaction satisfaction space after chapter nine does chapter 10 get us where we want to be as we then complete the book yeah Ali and then I knew what else actually chapter 10 uh is a pretty interesting chapter uh with all its um comparative analysis provided um and comparing active inference framework from with other um similar methodologies and unsurprisingly emphasizing uh almost always the advantages of using active inference over all the other Frameworks uh but yeah it is comparative analyzes uh were really helpful in my opinion in order to distinguish what exactly active inference is and how it relates to other uh other disciplinary areas but uh and I think that some of these materials could have been introduced much earlier I mean even in chapters uh one or two because uh most uh other some of the confusions at least some sometimes uh arises from not being too clear enough about what actor active inference is and what it does and how it compares to other related areas but otherwise it's a very very interesting chapter in my opinion I agree with that yeah I kind of feel slowly here that the way I would word it is that it's almost like a survey uh chapter but it's got a lot of details and it's like that would have been a good maybe not all of that up front in the same way but um it just helps delineate active inference a little bit more um uh just the textbook overall like I feel like uh we could make this no you know no exercises or something and lack of implementation comment it's like but uh the fact that it's in the structure of the way that the you know that it is currently understood at the uh Frontier is almost like a a little puzzle that will probably result in significantly better understanding of the domain and the you know the so I feel like it's uh it's just the right amount of challenge for me to like really ground what is being said here and it's not like just given to you so um usually that's not done very well once you have some pedagogical sort of

resource it's like this is what it is don't think about it just remember that and that's not very helpful either I wish there were maybe other libraries or something of software that explained a little more how this was working in a cogent fashion but also there's an opportunity to help write this so that's to me I don't know more interesting than just uh wrote definition explanation yes this isn't the final version of the textbook nor the final textbook ecosystem and I agree yes chapter 10 provides a nice overview it's it is long and it also may make sense as like chapter 1.1 even just to understand like the unifying theory of sentient Behavior angle before even talking about how to drive to active inference with a high road in the low road you know here's the overview in chapter one and here's Maybe here's how Hawaii is different than this other destination now here are two ways to get there now here's the generative model and empirical examples so many ways to read it Eric any overall thoughts or let's look at their key points um I I feel like this is kind of a kitchen sink chapter showing you know trying to take just a whole conglomeration of perspectives and lines of of research and ideas in the uh the whole Enterprise of computational intelligence cognitive science neuroscience some psychology and so forth and say okay this is how we view it through the active inference lens so I think that's the purpose kind of the purpose of the the chapters uh the second at least the second part of the chapter the first part of the chapter is kind of the summary of uh the again the overview of what active inference um is about and tries to tie the threads together and I don't I don't think it it's it they're tied any I mean it's good to see the reminders of you know what what are the what the journey has been through the book uh I appreciated that refresh um there are a number of you know threads that go in different directions um things like you know the notion of everything being a form of inference but um planning is a different form of inference because it's a different is a different functional energy functional than pure perceptual inference the notion of the exploitation exploration trade-off the notion of not being able to solve for the the base you know optimal policy but um therefore having to use um a variational energy approximation that was mentioned but I feel felt like the book never in this chapter didn't say well how do we really leverage that I mean how does that really come into play um you know operationally it um reviewed the examples that were used earlier in the book such as the team A's and so forth which I found underwhelming all the way through and um you know so you know the the chapter ends with promises it says wow this is this wonderful Hammer we have let's see what nails we can find out in the world and we're going to Hammer Hammer Hammer and let's see if they behave like nails you know we're going to treat them like they behave like nails even if they're screws um and you need a different

instrument to turn them so I think that as far as you know what if you're a devotee of active inference then this chapter is perhaps inspirational um if you're a skeptic and it's um it's like well okay okay that's nice move on yeah a lot there too let's let's look at some of these strong claims one benefit is that it provides of active inference is that it provides a complete solution to the Adaptive problems that sentient organisms have to solve of course more could be set but does it provide a complete solution or maybe a little bit still generously but perhaps less excessively it provides a complete framing or accounting which is different than a solution for the problems this makes it sound like just you know install octave.py and the problem is solved which is obviously going to be situational depending on the generative model which is under no guarantee to be effective just like no linear model is effective by principle so pretty obvious point but something where as Eric just suggested like a um adherent might see this as a slam dunk a fractal slam dunk whereas somebody who's approaching this more skeptically is actually not going to have seen a solution at least presented in this book it offers a unified perspective on problems like perception action selection attention emotional regulation which are usually treated in isolation in Psych and neuro and addressed using distinct computational approaches and artificial intelligence this much more coherently is true because it is a unified perspective on perception learning and but through the composability of generative models so merely the particular physics the Markov blanket itself is not a unified model of perception action selection attention and motion regulation Etc et cetera Etc however it's a composable framework under which these phenomena can be explored but just the active inference kernel figure 1.1 is not that but it does offer a perspective that allows us to integrate it in a way that linear models aren't going to get us there and indeed these fields do work in isolation across multiple levels of Mars hierarchy and so a unified perspective is really important to start having some discourse about like hey there are two computational functions from the same brain region or other kinds of um more integrative work so that's nice here they're going to contextualize active inference in light of the following other fields Eric first though go ahead yeah I'll just um add one thing since you mentioned Mar you know I I actually met David Maher once um when I was applying for for grad school and so I was a big part of the you know I was a big Enthusiast about the whole school of thought that he he was the founder of uh at MIT first in in the UK and then at MIT and one of the things that some of my fellow students and I observed was um in that era was a phenomenon we called physics Envy which is that um an observation that physics is the the grand science uh explained the universe and it manages to boil things down to very pithy equations simple equations that explain a lot you've

got E equals MC squared you've got Schroeder's equation um you know things like that and people in computer science and cognitive science and artificial intelligence want to have a grand Theory because we know that the even more interesting problem for civilization is not how the universe works it's how do minds work so we wanted it so at least our professors and our our mentors we're always looking for these Grand unified theories little equations that'll explain everything and it's important math is important so you know you can find like equations of um you know Optical flow and and that it's really important to model things um using math but um it seems that the free energy principle is is one of these um physics Envy type of approaches where we're looking for a single equation to explain everything and by the way it does borrow from statistical physics um so it really is and it I guess Tristan himself is his physical background so from that point of view um you know this is again you know looking for a hammer that will treat the mysteries of nature as a nail and you can hit it with a hammer and Presto you've got it and you know it may be more like biology where there are lots of principles there's lots of math to be doing lots of modeling you're doing but by golly it's really really complicated and complex and you're never going to boil it down to you know down the details of how it actually works and you know you can you know using any any simple equations physics does not predict um the shape of my coffee cup I don't care how much you know you know that that you understand children's equations or the laws of physics there's a lot other levels of complexity that have their own independent principles and um you know ways of modeling them that are required to really understand the phenomena so my concern is that these over encompassing claims about active inference are are falling susceptible to the the physics Envy track so sorry that's my oh awesome thank you yeah well it surely recalls the earlier um distinction of needs and scruffies from like chapter one in the book or I believe um whether the complexity of any realized system is so in the weeds and massively factorial that Transcendent generalizations are ineffectual um to the physics MV so this is the classic XKCD comic much memed many variants and um it also surely represents some sociological settings like at Santa Fe Institute and it's like well we can talk we can have the physicists speculate about sociology but you wouldn't have the sociologists speculate about physics right and there's cases where that conversation in both directions is helpful or not but it just isn't this projected down number line and then but as a sort of um rejoinder to the physics Envy is the fep carrying on this lineage of physics MV we may be in a lineage of physics annexation because the cognitive physics is actually expanding to tackle the scope of classical statistical thermodynamic and quantum physics under special cases of cognitive

systems and so I do agree that an individual might experience physics Envy or feel like they're getting closer and closer to the real thing as more and more equations and and inscrutable symbols appear however I wonder if on a deeper Supra personal level we're actually aligning physics more with cognitive science than vice versa which would completely invert the traditionally understood physics Envy they're going to have neuro Envy or they'll have they'll have mirmika Envy Ali yeah one other case in points in this uh physics Envy Endeavor is a Karen Broad uh who's a physicist which tries to apply her expertise and to uh and to some of the um cultural phenomena especially feminism and she has a very well-known book uh in that on that subject it's called meeting the universe halfway and she's also the founder of a kind of realism called agential realism uh which is a very uh isomorphic with some of the fundamental concepts of fep here as well so yeah there are lots of interminglings between uh Humanities and Sciences or especially physics these days yeah one image I kind of get is like there's a micro ratchet towards technicality applications visibility tangibility but there's a macro ratchet towards holism systems thinking Humanities and so on without broader awareness of that situation there can be Whiplash like are we doing philosophy on what things are and how they become and becoming an Unbecoming or are we calculating a particular form of a pomdp's expected free energy functional and we need to educate and develop our information environments to understand both and to include many perspectives yet on set so

10.2 wrap up just really quickly I'm going to also say there was an interesting discussion uh on uh like Lex Friedman's podcast with Andre carpathiana attention is all you need the way that they framed it this is just kind of this underwhelming kind of mimetic kind of pithy thing but they didn't make this claim about Transformers being kind of some significantly a more General uh trainable architecture that was just kind of in the paper slipped in and they were just framing it as here's better uh you know language uh translation slash prediction right and uh he didn't like that he was like no it was it was so uh transformational they should have named it you know that the Transformers the new whatever you know and then but then also had this conversation of well maybe it worked and it you know spread because it was this pithy meme thing that was under promising and then eventually over delivering I think in active inference I don't know I don't see like I see how this like uh fep um framing of it is like implicitly over promising but I don't see um a lot of people explicitly constantly kind of making this like fep is all you need active full explain everything sort of framing of it um but there's this I guess yeah and so like implicitly somehow because fep is this sort of physics framing of it it just it seems like uh it's murky because of that relationship but ultimately we'll

just find where it's uh Well Suited and and not that there aren't really in fact a lot of males out there and there may be a lot more screws or an equal amount of screws or or we don't know but we're going to get you know more complex and more precise models so how that exactly Works in implementation is Up For Debate but or it certain means to be seen But I I don't know if there's something that can be done there it's like that's already modeled as fep as a some sort of containing background foundational you know epistemic um prior there and so they've already said the you know Transformer is the amazing thing so it's just it will deliver until it meets that or and or not you know like you know yep nicely set like we have the worked example we have canonical neural networks our active inference agents we have the the initial tracers you know the grappling hooks have been thrown the bugle has been sounded now let's see what really happens so good comments 10.2 wrapping up oh I'll leave please uh sorry just one point about this claim in the book that says the active inference provides a complete solution to the Adaptive problems uh well I don't know it uh seems to me uh like a kind of dubious claim because uh especially about the word complete here uh maybe uh it they meant uh that to be an adequate solution to the Adaptive problems because uh you see there's a very uh significant and fundamental difference between being adequate and between complete in your Solutions right so in this case I'm not sure what they mean by a complete solution to the adapter problems because uh as we all know uh even at this stage of of its development active inference has some shortcomings and there are some uh well gaps in the theory and so on and uh so uh using this strong claim here uh seems to be uh a little bit out of place and uh yeah but uh I would be fine if it was if I have an adequate solution to some problems in some situations 10.2 might be a nice reading early it summarizes chapters summarizing chapter chapter one adaptive exchange the higher and the low road chapter two the low road Bayesian inference and generative models that can be inverted and heuristics thereof as well as decompositions associated with those heuristics chapter 3. the high road starting from the deflationary imperative to preserve integrity and avoid dissipation which we may have a more sophisticated realization of in the Bayesian physics chapter 5 neurobiology the most well-studied cases and systems of interest for many cortical subcortical motor architectures supporting the integrative work pointing to empirical examples on neurotransmitters composability of neural systems artificial systems that's part one of the book chapter six heads into part two of the book there's a recipe for Designing systems seven and eight are currently summarized as just the continuous and discrete time settings and chapter nine talks about model based data analysis in terms of parametric recovery of an individual's generative model

in the context of metabasian inference connecting the dots 10.3 done it and the whole iguana I guess usually we would say what the whole enchilada but that's not sentient enough so then it was out there it's a whole cognitive iguana it's a metabasian iguana we're imagining that we're building the whole iguana but we don't no one's actually built the whole iguana but also this is so representative how like threads of research just serve to be like like it's like spark after spark with no fire like what is it like to be a bat spawns tens of thousands of directions from 1954. but where can you go find what it's like to be a bat oh that's right it's still the exact same speculation that existed in 1954 which is if you had the motor apparatus and The evolutionary Heritage of being a bat then you'd know what it was like and it'd be whatever it is it is like like but if you're not you're not and so you don't know and I don't know if anyone has actually pierced that Veil so at least it reflects following up diligently or more diligently on provocations from our philosopher colleagues Eric yeah I'm sorry this is just a digression you sent me on well maybe the metaverse will tell us because then you can have your virtual reality environment that gives you only the abilities the sensory abilities and motor abilities of a bat and you live in that for a week and you learn really what about you know how to think like a bat what a bat can do can't do um I don't know just just sorry about that you no no no no because that's what I think is the brain machine interface like where it literally you'd rewire your brain to feel it yeah that and then then we'd know but you'd have to recover that too so there's like a difficulty of like bringing that sensory experience back to the human brain I don't all right well obviously yeah might be a no cloning theorem sort of situation there all right this is a close enough approach I'm gonna I'm gonna advertise a piece of work I did many years ago it's called the Consciousness machine and it's um it's a um it's an account for how to solve a hard problem of consciousness the hard problem being how do you uh not how does a why do we have Consciousness how does it work what how does it brain produce it but your own subjective experience of consciousness is um the mystery that it's hard to penetrate through pure science and explanations so how do you do that you have to have a user experience you basically have to make a it crossed between a video game and a simulation of your own mind the mind of your user and um in that environment then you can not only see intellectually well how is my brain working but you can feel it experience it and make it real to yourself and in the in the feedback sort of sense the motor feedback and so once we have a Consciousness machine which we don't know how to build yet but we're getting closer and a matter of uh you know all the the VR and technology is is an enabler for that then we'll be able to to um you know transform our conscious experience in various ways including uh becoming bats or ants

or or whatever by learning to live those kind of lives in certain contexts it's on my website awesome a short thought on that somebody will always say well you're not a bat you're a person thinking you're a bat but that's where you go to The Meta Turing test which is now your in the metaverse can you tell the difference between the true bat and the person thinking that they're about now if you can't design a setting to differentiate it from the functional ecological pragmatic View it says bat like in the metaverse space as you can get so yeah I mean active in the metaverse and internet of things and and the way that we can even just today qualitatively talk about cyber physical systems composability intelligence augmentation distributed cognitive systems these are all so valuable and they are not features that come from some of the competitors that we're going to be discussing today okay the whole iguana some connections to uh analyzes of cognition and generalized challenges in cognitive systems like exploration exploitation we talked earlier about how the planning mediated resolution of explore exploit was not quite that but certainly other papers do actually carry that out now that doesn't mean that they outperform any other given model but at least we know that we can frame explore exploit in a single step and in multi-step scenarios with active um and active offers a means of understanding corresponding neural computations the connection between cognitive processing and expected neurodynamics was actually very very interestingly characterized in the recent live stream number 51.1 which I highly recommend because this could be the damn breaking between artificial neural networks and active inference pomdp models so it's quite exciting to hear we will explore in the rest of this chapter some implications for psychological functions as if we were sketching a psychology textbook this is kind of like that girde Escher Bach style where it's like am I gonna end the book in the middle of the book and then fill it with filler I don't has that already happened I don't know am I sketching a book or is this the book itself that's up to you but that's such a funny note predictive brains Bayesian brain predictive coating predictive processing architecture [Music] these are popular Frameworks in computational cognitive neurosciences yet also at a deeper level these predictive approaches still struggle for recognition among somehow default non-predictive approaches like it always is needing to be justified in some people's eyes whereas active inference takes it several steps further first by incorporating action over multiple time skills although predictive processing via Andy Clark and others also was making directions in that way but in a less quantitative way and also introduces this variational physics grounding 4E cognition plus predictive processing with inference in action simply smashing those two together in your theory Collider would never reveal that classical statistical thermodynamic

and quantum mechanics can be understood as special cases of a cognitive physics so that's sort of this like third vector that is exceptionally technical but does more than or makes active inference more than merely for recognition plus predictive processing as qualitative syntheses with tractable implementation approach in the pomdp for example perception and learning we've talked about how there are similar inferential processes playing out over different time scales and that maps to some biologically plausible physiological updates in terms of synaptic plasticity for Learning and neural firing rates for perception and it's a very strong mapping indeed because the imperative in terms of the loss function of the neural network being trained includes the firing and the plasticity term and that actually brings a very strong synthesis between the loss function in the neural networks training and the variational free energy descents in the pomdp figure one in isomera at all and the loss function on a neural network as a function of observations and parameterization is concordance with the variational free energy in terms of observations and parameterizations we can model one interesting piece here I know this is not the textbook itself but this fictive causality causality is like the opposite of the expected free energy so it's almost like variational free energy is the now casting expected free energy is anticipated decision making and fictive causality is looking only at the past and it has a computational implementation in terms of a risk variable and a neuromodulatory interpretation with respect to dopamine modulation so very interesting and um tractable Bayesian brain I don't know what more can be said it's about time that the Bayesian brain got formal but there's a lot of Pros on the and a lot of reprinting of the Bayes equation but not much in the way of meaningful contributions beyond that and where there are meaningful contributions they're not carried out in the spirit of the unified sentient behavioral model it's just like oh this brain region can be understood as doing prior updating on this inputs so it's using Bayesian framework Bayesian statistics to describe neurophysiology but that is unless superintentionally otherwise on the scruffy end of the spectrum whereas active is like coming from the neat end of the spectrum and connecting some of those scruffy findings um action both action processing and perceptual processing are Guided by forward predictions many contact points with various motor embodied embedded perspectives that we've discussed ecological psychology affordances cups are for grasping change the world change your mind we're talking about action the consequences of past action what to do now planning in the future idea motor theory that's an interesting thread with William James and um Beyond and it is about connecting ideas with behavior that is getting formalized here cybernetics um the tote as well as the UDA Loop observe Orient decide Act and the test operate test exit model and

cybernetics brings in the concept of the goal-oriented uh telios and in certain cybernetic settings that was operationalized further as perceptual control theory which is that what is being bounded is surprise about perception interestingly and as um Dr isamara described in our conversation the Bayesian risk is calculated about uncertainty on hidden States which allows the Bayesian risk to be fit monotonically and with a um heuristic approximation and in doing so prevents overfitting of observables so in the fully observable chessboard situation of course observations in Hidden states are one of the same and the a matrix is just an identity Matrix mixing our metaphors whereas when there's uncertainty between the hidden State and the outcome one may be down a fallacious cul-de-sac if they overfit on the observations because there might be like Amanda mapping or many many mapping or stochastic mapping between hidden States and observations however under an appropriately structured partition the particular partition in particular one may fit a central tendency and variance attribute on an unobserved state and by doing so resists overfitting categorically to outcomes and also enables fitting of large or sparse data sets of observables optimal control theory discussed in various previous live streams and cases how our inverse and forward control theoretic in the control theory ontology how does it map to the way that we talk about um message passing and active inference General key point the cost or value function is um a special case or part of the decomposition of a variational or expected free energy however we're not shoehorning everything into cost or value however again components or special scenarios of the free energy can be understood to have a pragmatic value like or a cost like components utility and decision making dynamic programming and the Bellman work some of these ad hoc bonuses novelty bonus intrinsic reward and so on are being approached in a first principle's way through unified imperatives like the expected and variational free energy which can be decomposed into pragmatic and epistemic so whence utility oh utility where art thy sting how am I going to get utility by learning about active inference Bayesian decision making coming closer from the cybernetic and utilitarian decision-making space into specifically the Bayesian brain and Bayesian decision-making space this Bears very very heavily on the complete class theorem and so Ali thanks a lot for um raising this in the um preparations for 51. because the Bayesian beliefs can be framed in terms of willingness to bet willingness to pay but importantly from the complete class theorem perspective as basically like an actionability and the specific set of strategies that they are interested in and restrict their analysis to are the admissible decision rules which means that that decision rule is as good as other decision rules everywhere or at least somewhere so by staying in subspaces of admissible rules one has

strategies that are in some way meaning more functional than alternatives which is very interesting because that is still a large space with many bizarre strategies some of which are very fragile but their admissibility is a constraint that is equivalent to saying that the calculation of perception is going to be Bayes optimal like there's some signal input for which the common filter of time Horizon 3 7 11 is going to be the the um Bayes optimal solution or for whatever length of Kalman filter or Bayesian filtering approach you're taking you can calculate it Bayes optimally and that's in the inference space and then analogously we can think about the admissible strategies in the strategy space some reinforcement learning addressed but subtle reinforcement and reward are decompositions of expected and variational free Energies it's not that we don't reinforce high value behaviors it's just that the model architecture is quite distinct they then go into these families of RL model free reinforcement learning model based and policy gradients some similarities and differences with all of these planning is inference which we may or may not have seen in the book depending on how you think about it but the key point is that as future outcomes haven't happened even hypothetically one has to take a different approach towards planning and the consequences of hypothesized consequences of potential actions or policies it's a slightly different topic or question than inferring the relationship between actions taken in the world and the consequences on how things change and it actually is captured very interestingly again with the um fictive causality which can be understood as like a retrospective learning from just starting with the first data point already you're in the game but starting from even the first round and then accumulating through training learning the relationship of actions transitions in the world and allowing re-evaluation thereof because some actions have delayed consequences and so on learning that in an open-ended way while actions truly do have consequence in the real world in the generative process so active inference takes a unique approach for planning first we use variational inference second it's model based planning third we use an integrated expected free energy functional that subsumes other schemes like KL optimal control so out at worst it should fit these other schemes as well as they do albeit with some computational overhead at best we should strictly be better behavior and bounded rationality Cicero bounded rationality and free energy interestingly free energy Theory more of the psychology valence emotion motivation Dante homeostasis allostasis interoceptive processing we just completed 50 1.2 yesterday interception as modeling allostasis as control this is an important area and these models are fit attention salience and epistemic Dynamics so we talked we saw a kind of like utility oriented above with the bounded rationality and the reinforcement learning and so

on and now we're moving towards the epistemic components being addressed in a first principle's way um salience and attention now into the structural learning rule learning causal inference fast generalization building on the Bayes graph causal architectures and thinking about inference and inferring fast and slow meta learning active inference and other fields open active inference can be applied to other domains two domains social cultural Dynamics and ML and robots social and cultural thinking through other Minds the anticipating brain is not a scientist ecological cognitive processes Birdsong which as we've discussed isn't also the communication example that people think it is it's an example of diatically embodied pre-scripted monologue that's why it's called a duet for one this is not open-ended symbolic message passing so much much more needs to come into play before we can say that we have active inference models of like improvising entities but some of the turn taking architecture is adequate and for example these birds expect to hear a certain song but we could use this model to say well we just expect there to be a conversation we expect little dead air so we can use this mechanics to simulate the Dynamics of turn taking but we need a different payload for the actual communication if we want those entities those birds to be doing anything other than just saying what they already expect each other to say and in the machine and cumulative culture and all of these complexities extended cognition machine learning and Robotics JF is the one to speak to on this topic and it's really interesting with the cognitive robotics angle how like it's helping get action fully into the loop and by doing so showing which aspects of the inference and action are embodied embedded and cultured Etc et cetera Etc and we end with the Lord of the Rings quote perhaps referencing the one ring on the cover of the textbook perhaps not they started by daring to be naive and ask whether it is possible to understand brain and behavior from first principles and brought active inference from off stage to address that question they hope expect and prefer that the reader has been convinced that the answer to the original question is yes it is possible to understand brain and behavior from first principles the book is a useful compliment to philosophy and physics Kristin a physics for a particular systems 2019 monograph currently under contract at the same MIT press and speculatively we can say that this will be the active inference textbook and there is forthcoming a physics textbook that will be more about the fep the journey ends it was an introduction you can't just learn it in theory in theoretical neurobiology you have to get your hands on the models and their misbehaviors so like the classic Turing quip when someone said how can computers be creative when they're only put out what we put in they never surprise us we're just getting you know we're just getting what we put in we hear that today entering said I'm surprised by

computers every day whether or not you're going to be modeling computationally you may reflect on it day to day it might be about where you look it might be about where you lick it might be about what you do ultimately we are confident that you will continue to pursue active inference in some form Bravo to those who stay through the cohort do you have any last thoughts for today yeah well excellent reading thank you thank you for going through it in such detail that's a great summary of your part I'm trying to I'm trying to you know balance the um the epistemic and the pragmatic which is to say the comedic value we'll talk again about chapter 10 maybe go into a few more details and then probably highlight projects and the ways that we can return in 2023 for cohort three with something like a cleaner slate as we welcome the next set of people in we accreted a lot we got all the equations in we got all the figures in all the boxes are in now we have the the actual task laid out in terms of so we'll be ready to proceed all right going to finish the recording now thank you guys